

AMANORSAC STUDIO MIXING SUITE — One-Page Brief

Internal product-analysis document for the Amanorsac Studio standards program. Documentation only; repository state as of 2026-09-02 is the source of truth. Version 0.1.1 · Windows x64 · JUCE 8.0.15 · Status: **Release Candidate (internal)**

3 · MAIN INTERFACE (EXISTING APPROVED DESIGN REFERENCE)



assets/designs/D01.png — PRISM EQ (D01), the suite's flagship. Reused as-is; no new screenshot captured. In the running product D01's editor is a native JUCE re-implementation of this reference (analyzer graph at 30 fps, band strip, control dock). The other 20 products embed their own reference PNG and draw it directly, adding a live parameter-inspector panel on the right. Analog products (A01–A10, R01) follow a dark-metal/brass/cream hardware look with VU-style meters.

4 · VISUAL / UX IDENTITY (AS IMPLEMENTED)

Primary	#03070C digital bg · #08101A panel #0F0D0B analog bg · #181B19 metal
Accents	#2E7EFF electric blue (digital) · #D89D3F brass (analog) · #F0E5C8 cream D01 adds cyan/teal and a 6-colour band palette
Text	#F4F7FA primary, #A9B3C2 secondary; uppercase, letter-spaced labels
Surfaces	Near-black, subtle linear gradients, 1 px hairline borders, 7–9 px rounded panels; analog series uses dark metal + brass hairlines
Typography	No custom typeface: JUCE default sans (Segoe UI on Windows), bold 8–27 pt scale
Knobs	JUCE rotary (drag), accent-coloured arc, value box below, double-click resets, Shift = fine wheel. Faders: horizontal sliders in D01 compact strips only
Buttons / toggles	Stock JUCE TextButton / ToggleButton recoloured (panel fill, accent text/tick)
Dropdowns / tabs	JUCE ComboBox (panel bg, accent outline). No tab component; paged "control pages" with prev/next + selector. Cards: rounded parameter cards in a 2-column inspector grid; no menus or header bar outside D01
Meters / visual	D01: live spectrum + EQ curve graph, 30 fps timer. Others: none live (meters exist only in reference art)
States	Hover/pressed: JUCE LookAndFeel_V4 defaults. Disabled: nav buttons greyed at page limits; D01 inactive bands dimmed to 25–35 % alpha. No custom focus styling
Motion	None beyond the D01 graph refresh; no transitions or easing
Character	Dark, cinematic, "pro-audio hardware" references vs. minimal functional runtime chrome; the gap between the two is the key design observation

1 · PRODUCT IDENTITY

Name	Amanorsac Studio Mixing Suite (20 products + ANALOG MIX RACK)
Category	VST3 audio-effect plugins, each also shipped as a Standalone desktop app
One sentence	A 21-product mixing suite of ten digital processors (D01–D10) and ten analog-modelled processors (A01–A10), plus a fixed analog-only rack (R01) chaining A01–A10.
Purpose	Give mix engineers a complete, consistently built EQ / dynamics / spatial toolset from one vendor.
Users	Mixing and mastering engineers, producers using VST3 DAWs on Windows.
Status	Release Candidate, internal. Full Release build, 28/28 automated tests, installer verified. Not yet scanned in a third-party DAW, unsigned, no calibration corpus (KI-0001...0004).

2 · WHAT IT DOES

Digital D01–D10	PRISM EQ (24-band precision EQ), FLUX dynamic EQ (12 bands, sidechain), SPECTRA multiband dynamics (min/linear phase), limiter, de-esser, resonance control, spectral shaper (6 zones), multi-tap delay (8 taps), algorithmic reverb, multiband stereo imager (5 bands).
Analog A01–A10	Analog EQ / preamp / console, tape and valve saturation, FET / opto / VCA compressors (A06 has external sidechain), passive EQ, plate reverb.
Rack R01	All ten analog cores in fixed order with per-module enables; same DSP as the standalone products.
Workflow	Insert plugin → adjust automatable parameters (knobs, toggles, choice menus) → state saved with the DAW session. No preset system yet.
Distinctive	Every product is generated from a JSON contract (stable numbered parameter IDs, fixed banks); one shared wrapper, per-product DSP; the rack reuses cores rather than duplicating them.

5 · TECHNICAL SNAPSHOT

Stack	C++ +20, JUCE 8.0.15 (FetchContent-pinned), CMake 3.25+, Visual Studio 2026 (MSVC), .NET 8 for the installer bootstrapper. JUCE modules only (audio_utils, dsp, gui_basics); no other third-party code
Platforms	Windows 10/11 x64 only today. No macOS/AU, no AAX (no SDK/signing), no Linux
Formats	VST3 (Fx category) + Standalone .exe for all 21 products; manufacturer code Amns
DSP architecture	ProcessorModule boundary: control-thread setters → atomic/smoothed storage; realtime process() is allocation/lock/log-free. AnchorDSP selects the algorithm by plugin ID; rack slots host the same modules. Latency reported (D03 linear phase 63 taps). Sidechain buses on D02/A06
State	JUCE AudioProcessorValueTreeState generated from specs/plugins/*.json; parameter IDs normalised (DEC-0002); binary state via get/setStateInformation
Build / QA	CMake presets, 28 CTest suites (DSP matrices 44.1–192 kHz, state recall, contract validation, rack), bundle audit, startup smoke
Auth / licensing	None. No accounts, activation, or copy protection
Installer / update	Self-contained per-user single-file EXE (payload zip + PowerShell scripts), Start Menu shortcuts, HKCU uninstall entry. Unsigned. No updater
External services	None at runtime (no network, curl and web browser disabled). Telemetry/usage reporting off

6 · AMANORSAC STANDARDIZATION NOTES

KEEP product-specific
- Digital (blue/cyan, holographic graph) vs. analog (brass/cream, metal, VU) series identities - D01's per-band colour language and live spectrum/EQ-curve graph - Product taglines and per-product reference artwork - Fixed numbered parameter banks specific to each processor
STANDARDIZE across products
- Header: logo lockup, product name/subtitle, undo/redo, A/B, help, menu (present in art, mostly absent in runtime) - Shared Theme tokens (bg, panel, border, accent, text) and a real Amanorsac LookAndFeel replacing JUCE defaults - Knob, toggle, combo, meter, card and page/tab components; hover/pressed/disabled/focus rules - Typeface choice and type scale; window size/resize limits (currently 1320×600 vs 1280×800 vs 1280×720) - Contract-driven build, ID naming (DEC-0005/0006), test matrix and installer pipeline as the engineering standard
MISSING company-level infrastructure
- Amanorsac account integration; licensing / activation - Preset system and standardized preset browser (art shows one; none implemented) - Auto-updater; code signing (Authenticode, later AAX/AU notarization) - About screen, version display, legal/privacy/EULA links - Accessibility (only basic title/description on controls), keyboard navigation, screen-reader review - Crash reporting and telemetry with user consent; centralized settings; localisation - Runtime meters/VU, A/B, undo/redo, oversampling UI shown in references but not built